

MATH 550: Homework 0

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1) A centered second-order difference was used at the interior points of the domain, while first-order forward and backward differences were used at the left and right boundaries, respectively, to approximate the first derivative of $f(x) = \exp(\sin x)$ on $[0, 2\pi]$.

The numerical approximation is compared with the exact derivative in Figure 1. The close agreement confirms that the finite-difference approximation captures the derivative well, although the comparison alone gives limited information about the error behavior.

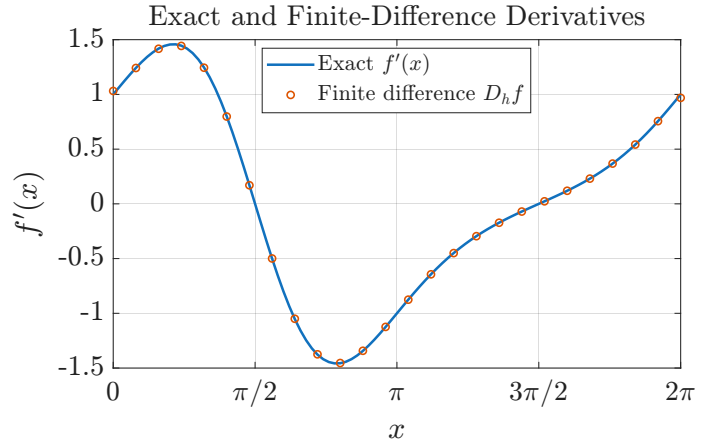
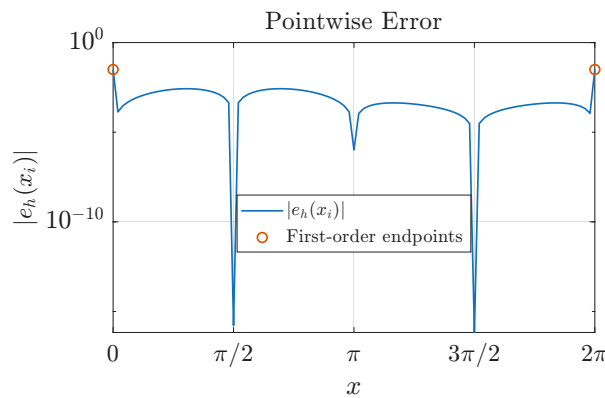
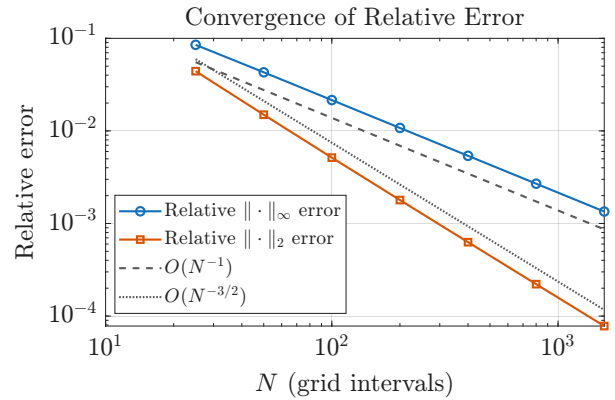


Figure 1: Exact and finite-difference derivatives for $N = 100$.



(a) Pointwise absolute error with the first-order endpoint errors highlighted.



(b) Relative error convergence in the infinity and 2-norms with reference lines.

Figure 2: Error behavior of the finite-difference approximation.

For the 2-norm, the error is dominated by the first-order endpoint errors, since the interior points are approximated with a second-order method. The error-vector norm therefore scales like $O(h)$, while the norm of the exact derivative grows like $\sqrt{N} = O(h^{-1/2})$. Thus, the relative 2-norm error scales like $O(h^{3/2}) = O(N^{-3/2})$. The infinity norm is dominated by the first-order endpoint errors and therefore scales like $O(N^{-1})$. These rates are confirmed in Figure 2b, where the reference slopes closely match the observed convergence rates.

2) The convenience of having our solution be periodic allows us to use centered second-order differences at all points in the domain, including the endpoints. The numerical approximation is compared with the exact derivative in Figure 3. We can see from the figure and included table in the code that the relative error terms for both norms converge at a rate of $O(N^{-2})$, which is consistent with the second-order accuracy of the centered difference approximation at not just the interior points but for the endpoints as well.

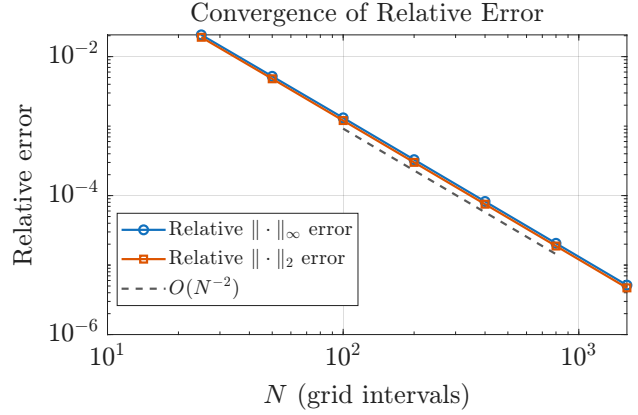
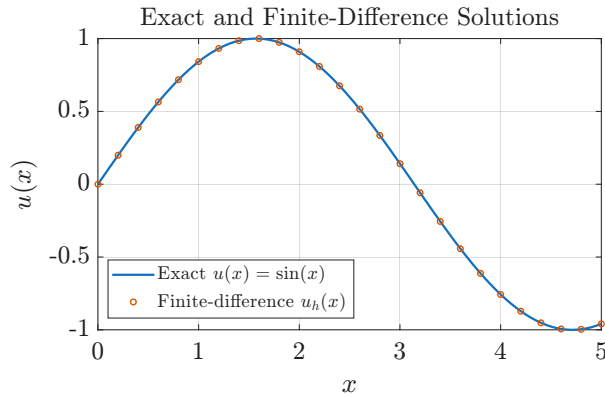
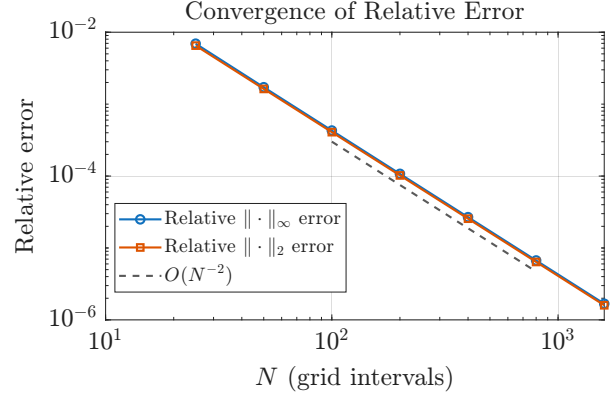


Figure 3: Infinity and 2-norm convergence for the periodic finite-difference approximations.

3) Our next problem considers a given ODE where we want to establish Dirichlet boundary conditions and a forcing function. Our method of manufactured solutions allows us to choose a solution and then determine the forcing function and boundary conditions that will produce the solution. We have chosen the solution $u(x) = \sin(x)$ and set the corresponding boundary conditions and forcing function to test the solution. The centered second-order approximations produce a tridiagonal linear system because each interior equation couples only neighboring grid values. The Dirichlet data are known, so their contributions were moved to the right-hand side before solving. The system was solved for interior points and then reconstructed. Our numerical solution is compared with the exact solution in Figure 4a. The relative error convergence is shown in Figure 4b, where we can see that the error converges at a rate of $O(N^{-2})$ for both norms.



(a) Solution comparison for $N = 100$.



(b) Relative error convergence in the infinity and 2-norms with reference lines.

Figure 4: Error behavior of the finite-difference approximation.

4) Solving the Poisson equation over $x \in [0, 5], y \in [0, 5]$ with Dirichlet boundary conditions, we used a second-order centered difference approximation for the Laplacian operator. The boundary conditions were explicitly declared, the interior was solved for, and the solution was reconstructed. The numerical solution was compared with the exact solution, and the error convergence table and plot constructed. The results confirmed that the finite-difference approximation approaches the exact solution with a convergence rate of $O(N^{-2})$ for the infinity norm, consistent with the second-order accuracy of the method.

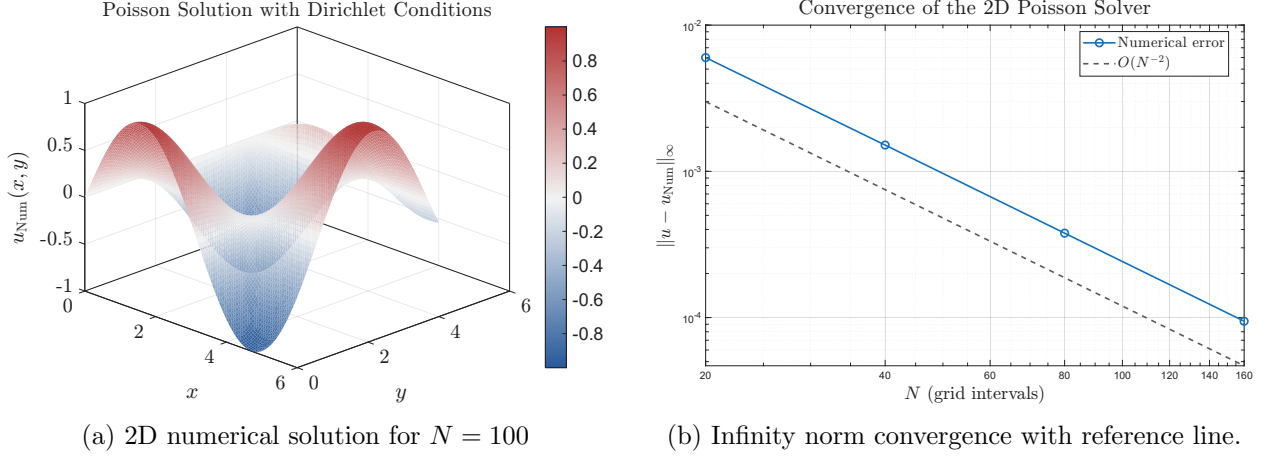


Figure 5: Error behavior of the finite-difference approximation.

With Neumann conditions imposed on all four boundaries, the resulting linear system is singular. In the discrete system, the constant vector lies in the nullspace of the finite-difference Laplacian, so the solution is only determined up to an additive constant. The same issue occurs in the PDE: the Neumann data prescribe derivatives of u , not its absolute level, so $u + C$ satisfies the same boundary conditions. A unique numerical solution can be obtained by adding one normalization condition, such as fixing $u(0, 0) = 0$ or prescribing the mean value of u .